

Bearing Measurements

Odometry

Local Mapping

Effective-Observation-Aware DRLS-based Estimation

Range + FoV + LoS Checks
Effective-Observation Indicator

Pseudo-Linear
Observation

Dynamic
Prior

Distributed Information Fusion
Local / Neighbor Effective Information

Posterior Recursion
Target Estimate State

Probabilistic-Observability-Driven Active Sensing Configuration

Planning Belief

Shared Target
Motion Scenarios

Future Effective-
Observation Prediction

Local Scenario Information

Cross-Agent / Cross-Time
Information Aggregation

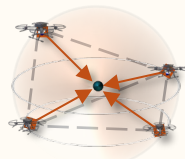
Normalized Joint Information Matrix

Information Utility

Cooperative Sufficient-
Observability Probability

Configuration
Repulsion Cost

Projected
Gradient Ascent



Goal Viewpoint

Motion-Primitive Trajectory Planning

Offline
Generation

Path $\rightarrow$ Time Parameterization
 $\rightarrow$ Motion-Primitive Library

Online Planning

State + Goal Viewpoint + Environment
+ Predicted Target + Neighbor Trajectory

Hard
Rejection

Obstacle Collision

Flight Area

Primitive
Cost

Target Proximity

Inter-UAV Distance

Goal Viewpoint

Primitive Selection $\rightarrow$ Yaw Planning $\rightarrow$
Receding-Horizon Execution

Communication
Network

Pseudo-Linear Measurements;
Predicted States

Information Matrix;
Viewing Directions

Selected Trajectory

UAV Controller